

Part A Chapter 2

Growth and Medicine Decay¹

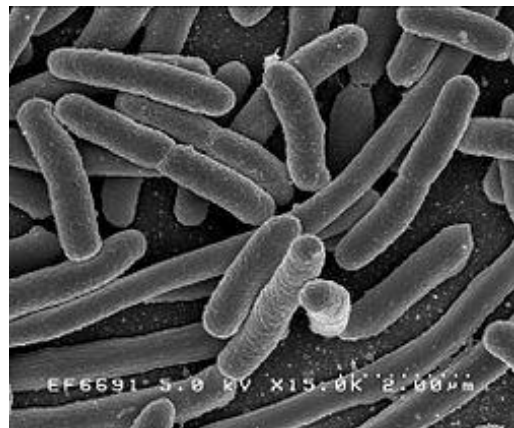
1 Introduction

Growth can be used to refer to biological growth processes, but it can also be used in a more metaphorical sense to indicate increase of some quantity. For example, the economy can be expected to grow (and it can also be reported to actually have grown), companies are proudly announced to be growing, and even trust in somebody or a relationship can be said to have grown. In this chapter, first processes in biological context are taken as a point of departure, but the models developed will be of a more generic nature and will also apply to cases where the growing happens more metaphorically.

1.1 Domain description and focus

In biological context growth can be found in various variants. Organisms, such as human beings can grow, but also, for example, plants, tissue with cells dividing in an uncontrolled manner such as tumours, and populations of a certain specie, such as a population of fish in the sea, or a population of infectious bacteria in someone's body. All these biological growth processes originate in growth at the cellular level. Roughly spoken a cell can support growth by two mechanisms: by enlarging its dimensions and by cell division. As the former mechanism usually plays a modest role in growth processes as an independent mechanism, most biological growth crucially depends on cell division (in combination with enlarging the generated cells to the original size of the parent cell).

The effect of cell division on the overall growth depends on where the cell division takes place. Sometimes all cells keep on dividing; for example this happens in a population of bacteria. In other cases, however, only some of the cells (continue to) divide, for example at the tip of the stem of a plant or of the root of a plant. This affects how fast the overall growing takes place.



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1.2 Characteristic patterns in growth phenomenon

A number of often occurring patterns of growth can be distinguished. In this chapter linear growth, exponential growth (positive and negative), and logistic growth will be addressed. The different variants of growth will be discussed below. When the model is evaluated, it will be checked whether the model is able to exhibit these patterns.

Exponential growth

As an example, in Figure 1 the case is shown where after a fixed time interval (the time duration needed for a division) every cell divides again, which again and again leads to a doubling of the whole population during such a time interval: *exponential growth*. The time duration after which an amount has doubled is called the *doubling-time*.

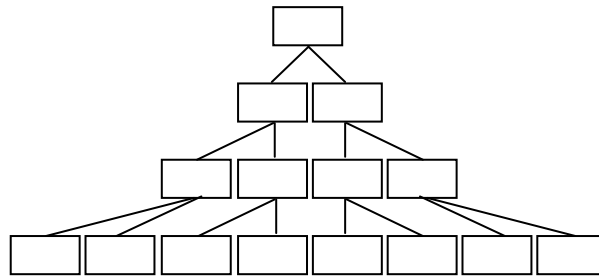


Figure 1 Exponential growth by cell division everywhere

Linear growth

In contrast, in Figure 2 a growth process is shown where only the cells at the tips divide: as soon as a cell is not a tip cell anymore, it stops dividing. For the whole population this leads to the addition of a constant number (two in this case) of extra cells each time interval of the duration of one division: *linear growth*.

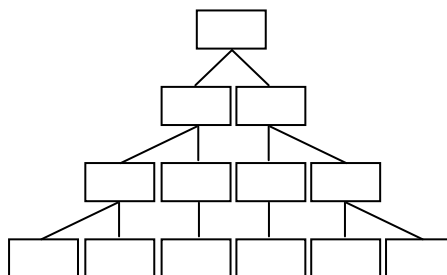


Figure 2 Linear growth by local cell division

Limited growth: sigmoid or logistic growth

Another important distinction is that between *unlimited growth* and *limited growth*. In many cases environmental or control factors limit the growth. For example, the environment may provide only a limited amount of food for the cells, so that the more cells there are, the less growth occurs (such a limitation is sometimes called the *carrying capacity* of the environment, indicating how many organisms the environment can support or carry). Moreover, within an organism often a sophisticated system of hormones controls growth by limiting it. Therefore the most often occurring growth pattern in nature is first of an exponential or linear type, but after some initial period the growth becomes less fast, and at the end stops completely. As the graph of such a pattern looks a bit like the shape of a large *S* (e.g., see Figure 5 below), it is usually called a *sigmoid* (after the Greek letter sigma) or *logistic growth* pattern (not to be confused with logarithmic patterns, which are different; they have no limit).

Negative growth: negative exponential decay

It may also be the case that in each period not only new cells are generated, but also cells may disappear because they are dying or are taken away. This can happen because there is not enough food to live, or poison in the environment (for example, antibiotics), or even killer cells that eat them. This provides a kind of opposite process, which may cause a population of cells to decrease instead of increase. In this case for each period a *gain* occurs in the sense of extra cells generated, and at the same time a *loss* occurs in the sense of cells that have disappeared. If the gain is more than the loss, the population still will increase, in the other case it will decrease in size.

Case study: management of medicine usage

A medicine in the body interacts with other substances that are present, which makes the level of the medicine decrease and the medicine disappear after some time. The number of these interactions per time unit is proportional with the number of available medicine molecules. For example, if there are twice as many molecules, then twice as many interactions occur. Therefore every time unit a fixed fraction of the medicine disappears. This makes that the medicine level shows a negative exponential pattern. For each medicine it is known how fast this process goes. The *half-life time* of a medicine indicates the (fixed) time duration after which an amount (e.g., concentration in the blood) of the medicine has halved. Medicines with a short half-life time have to be taken often, and the level can strongly fluctuate. Medicines with longer half-life times are easier in use.

2 Analysis of the Main Aspects of Growth and Their Relations

This section describes globally the relevant concepts used for the analysis and the model, and the relations between these concepts.

2.1 Identification of relevant concepts

To describe growth processes as discussed above, a number of concepts can be used. In the first place a certain *amount* (or number) is considered: that what grows. Next, the amount (per time unit) by which it grows, the *gain* and the amount by which it decays, the *loss*, are considered. These amounts are related to a *gain factor* and a *loss factor*. A gain factor is a relative indication: it indicates the gain per unit of the amount. So, the total gain can be obtained from the gain factor by multiplying it by the amount. For example, if in a population of 200 individuals, every individual gets two children, then the gain is 400, and the gain factor is 2. Similarly the loss factor indicates relative loss. If in a population of 200 a loss of 50 occurs, then the loss factor is $50/200 = 0.25$. Notice that if there is only gain and no loss, then the loss factor is 0. Similarly if there is only loss and no gain, then the gain factor is 0. In summary, the following five concepts are used:

- amount
- gain and loss (per time unit)
- gain and loss factor

2.2 Relationships between the identified concepts

The dynamic relationships between the identified concepts are as follows. For a graphical overview, see Figure 3.

- The amount and the loss factor affect loss
- The amount and the gain factor affect gain
- The gain and loss affect the amount
- The amount affects the gain and loss factor

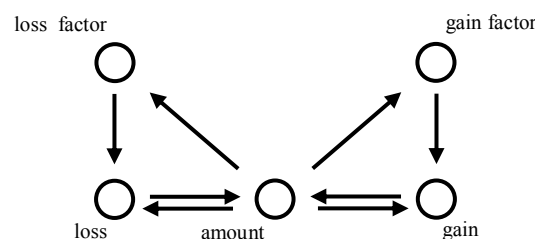


Figure 3 Dynamic relationships between the concepts

3 A Detailed Model

In this section more details of the model described here are discussed. First it is shown in which way the relevant concepts are formalised, and next the dynamic relations are formalised in a detailed manner.

3.1 Formalising the concepts used

To obtain a detailed model, first the concepts used are formalised. In this case for each of the concepts a numerical variable is introduced that has real numbers as values; see Table 1.

informal	formal
amount	<i>Amount</i>
gain (per time unit)	<i>Gain</i>
loss (per time unit)	<i>Loss</i>
gain factor	<i>GainFactor</i>
loss factor	<i>LossFactor</i>

Table 1 Formalisation of the relevant concepts by numerical variables

These concepts are used to describe aspects of states. As states differ for different points in time, in principle the values of these numerical variables depend on time. For example, if t is a time point considered, then $Amount(t)$ denotes the amount at that time point.

3.2 Formalising the dynamic relationships

The dynamics of the relevant concepts involve mutual interactions between them over time. These interactions can be expressed by relationships over time (*temporal relationships*), which are formalised in more detail below. They describe the basic parts of the process as shown in Figure 3. Such temporal relationships describe how the value of a certain variable at a time point t_2 , for example, $Amount(t_2)$ depends on values of variables at some earlier time point t_1 , for example $Loss(t_1)$ and $Gain(t_1)$. To get a systematic treatment, the time points t_1 and t_2 are taken to differ a fixed time duration Δt . For example, if time is measured in seconds, and Δt is 0.4 seconds, then $t_2 = t_1 + 0.4$. So, the general pattern is that temporal relationships are considered between values of variables in the state at $t + \Delta t$ and in the state at t . The time difference Δt can be used as the *step size* with which the model calculates the pattern over time.

Amount and gain factor affect gain

Gain is calculated by multiplying the gain factor by the amount:

$$Gain(t) = GainFactor(t) * Amount(t)$$

Amount and loss factor affect loss

Loss is calculated by multiplying the loss factor by the amount:

$$Loss(t) = LossFactor(t) * Amount(t)$$

From gain and loss to amount

Gain adds to the amount and loss subtracts from the amount, both taken over the time interval that is passing.

$$Amount(t + \Delta t) = Amount(t) + (Gain(t) - Loss(t)) * \Delta t$$

Amount affects gain and loss factor

In how far the amount affects the gain or loss factor depends on the type of growth that is considered. For example, focusing on a situation with only gain and no loss, the following cases often occur:

- The gain factor is independent of the amount; it is constant over time:
 $GainFactor(t) = \alpha$
 for a fixed α . This describes an unlimited exponential growth process; see also Section 4, Figure 4.
- The gain factor depends over time on the amount as follows. It is proportional to the relative difference between the amount and C for a constant C , so the gain factor decreases as the amount approaches the value C . Expressed in a formula, the gain factor is proportional to

$$\frac{C - Amount(t)}{C}$$

which gives the gain factor

$$GainFactor(t) = \alpha * \frac{C - Amount(t)}{C}$$

for a fixed α . This describes a sigmoid or logistic growth process, where growth is limited by C ; see also Section 4, Figure 5, where $C = 1000$. For loss factors, similar examples can be given.

In Table 2 the dependencies over time are shown for an example scenario.

t	loss factor	loss	amount	gain	gain factor
0	0.1	10	100	40	0.4
1	0.1		130		0.4
2	0.1				0.4
3	0.1				0.4
4	0.1				0.4

Table 2 Dependencies over time

4 Simulation Experiments

The model as described above has been used to make a number of simulations in Microsoft Excel. As a first example scenario, in Figure 4 an unlimited growth process with gain factor 0.4 and loss factor 0.1 is shown. Here and in the other pictures the horizontal axis shows the time. For this case:

$$\text{GainFactor}(t) = 0.4$$

$$\text{LossFactor}(t) = 0.1$$

Then

$$\begin{aligned} \text{Amount}(t + \Delta t) &= \text{Amount}(t) + (\text{GainFactor}(t) - \text{LossFactor}(t)) * \text{Amount}(t) \\ &= \text{Amount}(t) + (0.4 - 0.1) * \text{Amount}(t) \\ &= \text{Amount}(t) + 0.3 * \text{Amount}(t) \\ &= 1.3 * \text{Amount}(t) \end{aligned}$$

This shows that after every time interval of Δt the amount is multiplied by a factor 1.3. This can also be seen in the graph depicted in the upper part of Figure 4. Note that also for the gains and losses per time unit, an exponential pattern is shown (lower part of Figure 4).

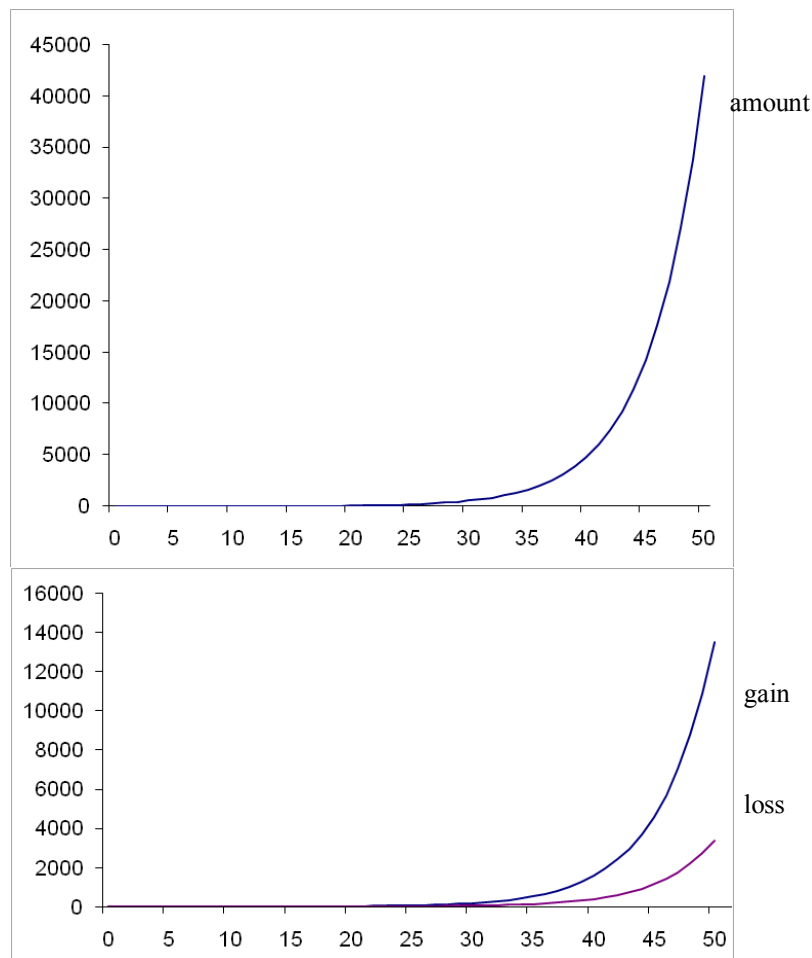


Figure 4 Exponential growth process with gain factor 0.4 and loss factor 0.1

Next a growth process is shown with a limitation of 1000. To incorporate this, the gain factor was set proportional to the (relative) difference of the amount with 1000.

$$\text{GainFactor}(t) = 0.3 * \frac{1000 - \text{Amount}(t)}{1000}$$

This time the loss factor has been put at 0. The pattern of the amount depicted in the upper part in Figure 5 shows a logistic (or sigmoid) trend, bounded by 1000 indeed. Notice that the gains (per time unit) (see the middle part in Figure 5) increase to a certain maximum at the point around 30, exactly at the point where the amount graph is most steep. The lowest part in Figure 5 shows the gain factor over time.

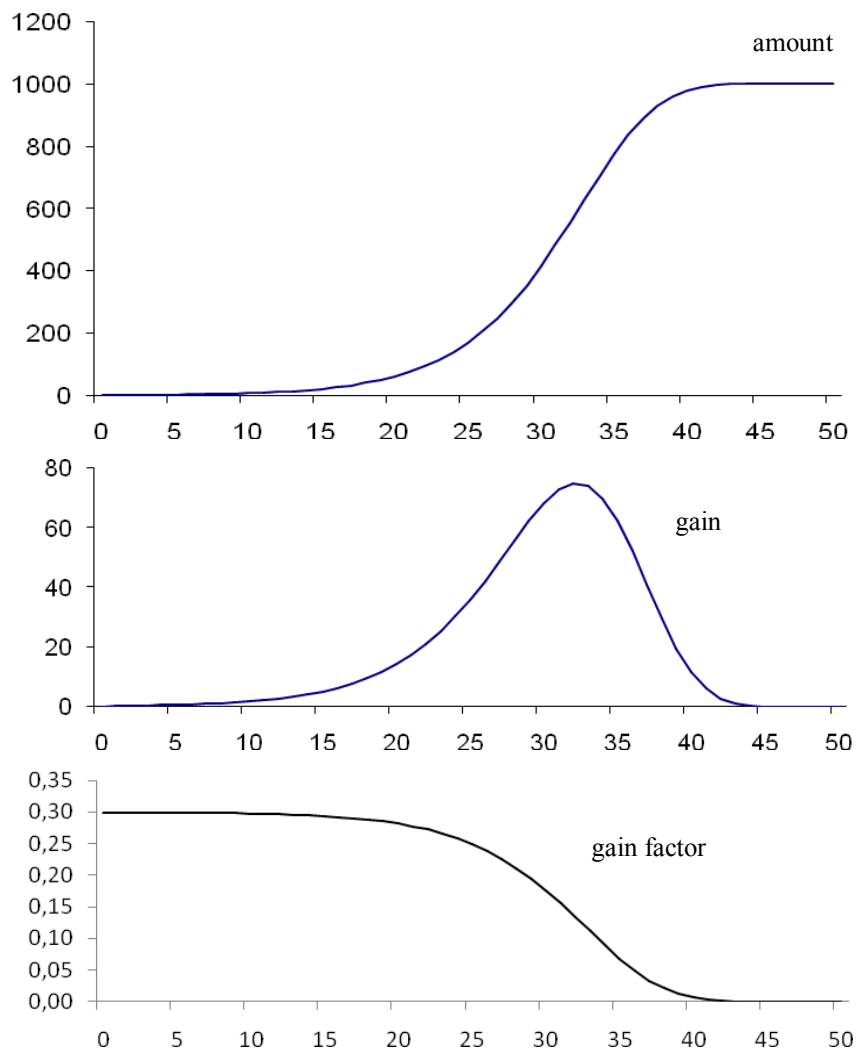


Figure 5 Example of a logistic growth process

In Figure 6 a situation is shown where the loss factor (0.4) is higher than the gain factor (0.35). The result is a negative balance for gain and loss, and therefore a negative exponential pattern.

$$\begin{aligned}
 \text{Amount}(t + \Delta t) &= \text{Amount}(t) + (\text{GainFactor}(t) - \text{LossFactor}(t)) * \text{Amount}(t) \\
 &= \text{Amount}(t) + (0.35 - 0.4) * \text{Amount}(t) \\
 &= \text{Amount}(t) - 0.05 * \text{Amount}(t) \\
 &= 0.95 * \text{Amount}(t)
 \end{aligned}$$

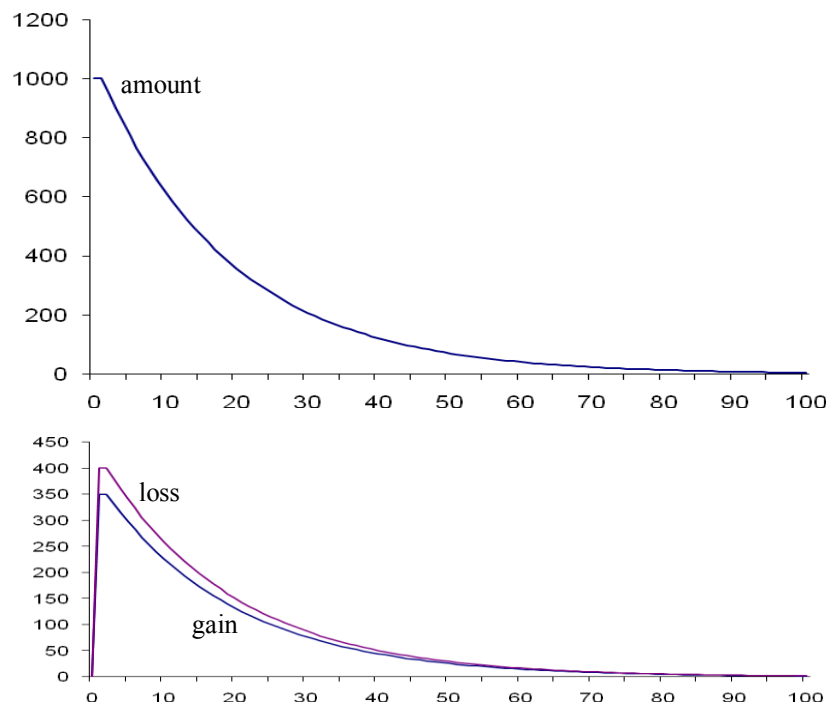


Figure 6 Negative exponential pattern due to higher loss than gain

Finally a case of medicine usage is shown, in Figure 7. Time is in hours. The loss factor chosen was 1.2, which corresponds to a half-life time of about 6 hours, as can be seen in the upper part of Figure 7 that shows the amount of medicine. For example, starting at 60 at time 0, six hours later, at 6.00 when the first pill is taken, the medicine level has dropped to 30. The gain factor was put on 0. Gains were modelled as taking a pill three times a day: at 6.00, at 13.00, and at 20.00. The pattern shown makes clear that serious fluctuations in medicine level easily occur (here between about 28 and 97, which is a factor of more than 3).

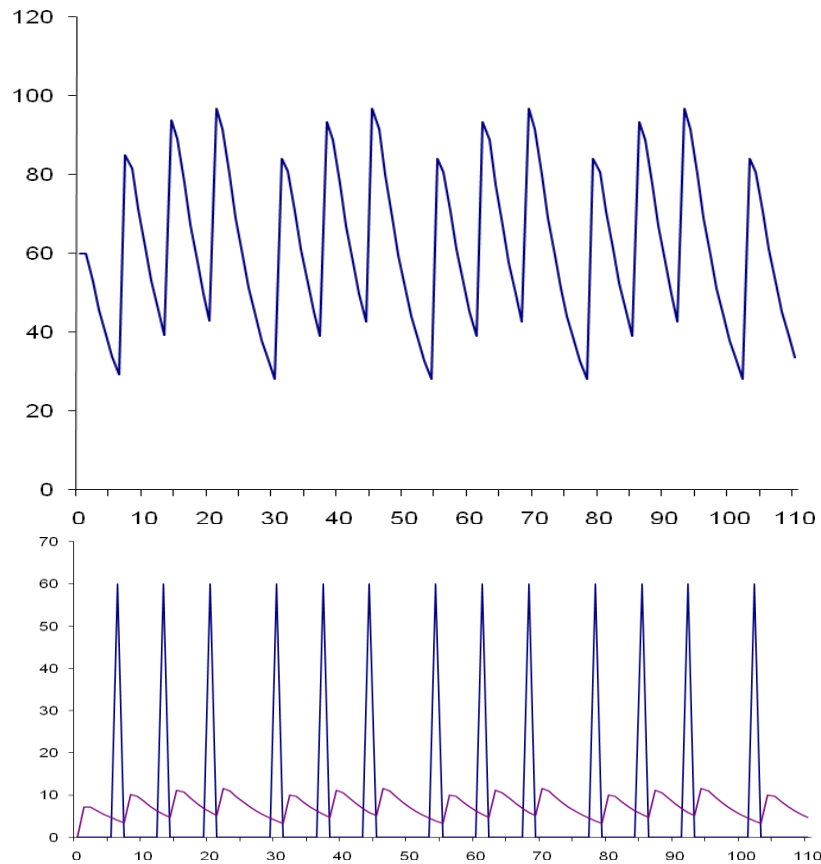


Figure 7 Fluctuations in medicine level

5 Evaluation

In this section, first the simulations performed using the model are evaluated with respect to the characteristic patterns identified in Section 1.2. Next, the assumptions behind the model are summarised. Finally, some variations and refinements of the model are discussed.

5.1 Evaluation of simulations and characteristic patterns

A number of different simulations conducted in Microsoft Excel show the characteristic patterns identified in Section 1.2. It is not difficult to perform some more experiments.

5.2 Assumptions behind the model

The assumptions that have been made for the medicine case include:

- Medicine taken is immediately in the blood
- Metabolism is always at the same level, hence a constant loss factor.

5.3 Possibilities for variation and refinement

A number of possible variations or refinements can be made for the medicine case.

- Try to get fluctuations less for the given half-life time
- Try the same with shorter or longer half-life times
- Explore how often medicine should be taken to have the fluctuations less than 20%.
- Incorporate differences in metabolism over time: in the night a lower loss factor due to lower metabolism (unless you are going out, or do a night shift, for example).
- Incorporate times to eat and the effect on the metabolism.
- Take into account a gradual uptake in the blood.
- Suppose the medicine has to fight an infection that basically can grow exponentially, for example, antibiotics, or virus brakers. What happens if one pill is forgotten?
- Incorporate that the medicine becomes less effective after some time.

6 Further Explorations

In this section three possible refinements of the model for medicine dynamics are briefly discussed. The first refinement concerns pills that slowly release their substance, the second fighting bacteria by antibiotics, and the third fighting viruses by virus brakers.

More advanced medicine models based on slow pills

As has been shown, by taking pills three time a day, the level of medicine can fluctuate by a factor or more. Solutions to this problem can be taking pills many times a day (which is a bit inconvenient) or having a constant flow of medicine directly in the blood, through an infusion (what is in fact only practical within the context of a hospital). Another solution that has been developed is to use pills that dissolve very slowly, so that the substance becomes available in a period of hours. This can be incorporated in the model to test the effect on the fluctuations in the medicine level.

Coupled models for bacterial infection and antibiotics

A second interesting extension can be made for the case the medicine is used to get rid of an infectious disease. Here the infectant (e.g., a bacterium) can be described by a model for its own mechanism to grow in number (in principle exponentially) and the medicine (e.g., antibiotics), by a model for its own negative exponential growth pattern. As the medicine has to limit the growth of the bacterium, the two growth patterns interact; the more medicine is available, the more the growth of the infectant is limited. On the basis of such a relationship the two growth models can be coupled. Fluctuations in the medicine level will have serious impact on the growth of the infectant: for example, as soon as one pill of antibiotics is

forgotten, the number of bacteria can explode again, undoing in one step the effects of the previous pills. Such patterns can be explored on the basis of the types of models as discussed in this chapter.

Coupled models for virus infection and virus brakers

Another theme that can be explored further is a virus infection and the use of virus brakers against it. Here the growth of the virus also depends on certain cells (the *target cells* or T-cells) of the host in which they reproduce. For example, in (Perelson and Nelson, 1999; Ellner and Guckenheimer, 2006, Section 6.6;) this is addressed for an HIV infection, and in (Baccam, Beauchemin, Macken, Hayden, and Perelson, 2006) for an influenza infection.

In this literature the idea is that the target cells have two ways of increasing in numbers per time unit:

- by a constant gain (from sources in the body such as the thymus) by which in principle linear growth can occur,
- by a gain based on a logistic growth pattern by cell division

Besides the number of target cells decreases per time unit

- by a constant loss factor

Moreover, per time unit

- a fraction of the target cells gets infected by the virus; this fraction is assumed proportional with the number of viruses.

Also these infected target cells decrease in numbers per time unit

- by a constant loss factor.

Furthermore, the virus increases in numbers per time unit

- by a virus gain based on infected cells that pass away, thereby producing a certain fixed amount of new viruses per dying cell.

Finally the virus decreases in number per time unit

- by a constant virus loss factor.

Virus brakers are used to block certain steps in the production of new infectious virus particles. Two possibilities are:

- a) Block the process to incorporate DNA originating from the virus within the DNA of the cell. This can be done by virus brakers inhibiting the enzymes reverse transcriptase (making the DNA copy of the virus RNA) and/or integrase (including the virus originating DNA in the cell's DNA).
- b) Make the viruses that are produced ineffective. This can be done by virus brakers inhibiting the enzyme viral protease (which makes the appropriate proteins available)

The effect of virus brakers of type a) is that

- from the fraction of target cells that may get infected by the virus only a small fraction is actually infected, depending on the effectiveness of the inhibiting virus braker.

The effect for a virus braker of type b) is that

- only a small fraction of the amount of viruses produced by a target cell passing away is viable, depending on the effectiveness of the inhibiting virus braker.

For more details, see the literature mentioned above, in particular (Perelson and Nelson, 1999).

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